

ANNEX W

*Maritime Telemetry & Dual-Key
Transit Protocol*

"Real-Time Verification for the Hormuz Transit Zone"

Self-Contained · Incorporable by Reference · Step-Back/Earnback Integrated

Version 1.5 — Integrated Stabilization Measure

Expires only upon mutual agreement or permanent IHTA charter supersession

MODULARITY-PRESERVING REVISION

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Annex W — Maritime Telemetry & Dual-Key Transit Protocol

Preamble

The Parties acknowledge that existing maritime annexes (Annex N.a, Stability First Phase II) establish political commitments for Hormuz transit but lack real-time telemetry verification, dual-key operational controls, and per-transaction financial mechanisms comparable to Annex V (Nuclear Dust Protocol).

No Party accepts unilateral verification authority over the Strait of Hormuz.

A multilateral, telemetry-first architecture provides continuous real-time monitoring of all transit activity, requires dual-key authorization for any change in operational posture, and links financial releases to verified, per-transit compliance.

The United States maintains its absolute position that the Strait of Hormuz is an international waterway where no unilateral tolls are imposed. Nothing in this Annex alters that position. All financial mechanisms described herein constitute releases of Iran's own frozen sovereign assets held in escrow — not payments from shippers, not tolls, and not new revenue.

The Islamic Republic of Iran maintains that the Persian Gulf Strait Authority (PGSA) exercises lawful sovereign authority over Iranian territorial waters. The United States does not formally recognize the PGSA as a legitimate tolling authority. This functional coordination mechanism is a sovereignty-preserving workaround, not a legal endorsement.

The Parties adopt this Annex W as the binding operational mechanism governing real-time verification, dual-key transit authorization, immediate conditional reopening, performance-based asset releases, automated breach triggers, and the Real-Time Compliance Dashboard for the Hormuz Transit Zone.

Present-Tense Execution Principle: Every clause in this Annex executes in the active present tense. Consequences activate in real time. No Party waits for subsequent authorization once a condition is met. Triggers fire automatically upon telemetry confirmation. Financial releases occur upon verification, not upon political consensus.

No Tolling Principle: No commercial vessel pays any fee, toll, or navigation service charge to any Party or to the JHA as a condition of transit. The zero-toll principle is absolute and non-waivable. All financial

releases under this Annex constitute performance-based returns of Iran's own frozen sovereign assets held in escrow.

Non-Recognition Clause: Nothing in this Annex constitutes recognition by either Party of the other Party's government, political system, or regional policies. The JHA is a purely functional administrative mechanism. The United States does not formally recognize the PGSA as a legitimate tolling authority. Iran's domestic legal arrangements are not affected by US non-recognition.

Step-Back Integration Principle: This Annex W incorporates the full Three-Tier Breach & Step-Back Architecture from the parent framework's Article 13, calibrated specifically for maritime telemetry violations. The tier system governs all micro-tranche releases, penalty assessments, and earnback procedures.

Modularity Clause: This Annex W is self-contained and incorporable into any existing or future agreement through a single sentence of adoption. In the event of inconsistency, Annex W governs with respect to maritime telemetry, dual-key transit authorization, immediate conditional reopening, performance-based asset releases, automated breach triggers, and the Real-Time Compliance Dashboard. The financial architecture within this Annex operates exclusively on maritime performance metrics. No land-based or non-maritime operational module exerts cross-module influence over the maritime escrow release schedule.

Article 1: Real-Time Telemetry Infrastructure

The Joint Hormuz Authority (JHA) Maritime Coordination Cell (MCC) in Muscat operates the following real-time telemetry systems for the Hormuz Transit Zone (HTZ). All data feeds transmit simultaneously to all Parties with identical latency.

Table W-1: Telemetry Infrastructure and Data Feeds

System	Function	Feed Recipients	Latency Requirement
Radar Network (3 stations)	Continuous surface tracking of all HTZ vessels (military and civilian)	JHA MCC, Iran (PGSA/IRGC-Navy), US (5th Fleet), Oman, IAEA	Real-time (5-second refresh)
Cryptographic AIS Verification	Tamper-proof "JHA-AUTHORIZED" status with rolling code authentication	All Parties; Lloyd's List; UN OCHA; IMO	1-minute position updates

Seismic/Acoustic Array	Mine detonation, underwater drone, submarine, or unauthorized tunneling detection	JHA MCC; Iranian Navy; US 5th Fleet; Oman	Real-time alert (<30 seconds)
Environmental Sensor Grid	Oil spill, chemical, radiation, or biological hazard detection	Iran EPA; IAEA; JHA; UNEP	15-minute sampling cycles
Transit Mass Balance	Real-time count of vessels, tonnage, and compliance status	All Parties; Shared Digital Registry (Annex U)	Updated every 15 minutes
Encrypted Video Network	Visual monitoring of the Commercial Transit Corridor (CTC)	All Parties; Joint Secretariat; JHA Council	Live (15-second max interruption)

Data Sharing Symmetry: No Party receives preferential access, earlier notification, or exclusive data feeds. Any data asymmetry constitutes an automatic Tier 2 Material Disruption under this Annex W.

Article 2: Dual-Key Authorization for Transit Operations

Every change in HTZ operational posture and every issuance of a JHA-AUTHORIZED transit permit requires two digital keys operating in tandem.

2.1 Digital Key Architecture

Table W-2: Digital Key Holders and Authorization Functions

Key Holder	Key Type	Authorization Function
Islamic Republic of Iran (PGSA/IRGC-Navy duty officer embedded at JHA MCC)	Iran Key	Certifies that no Iranian forces are conducting prohibited interdiction under Annex N.a; confirms stand-off distances are maintained
JHA Maritime Coordination Cell (Omani Chair, with IAEA verification observer)	JHA Key	Certifies that all telemetry systems are operational, no unresolved breach exists, and verification standards are met

2.2 PGSA Operational Integration (Sovereignty-Preserving)

The Persian Gulf Strait Authority (PGSA) operates as the designated Iranian coastal enforcement body within Iranian territorial waters (0–12nm). This designation is a factual acknowledgment of existing Iranian administrative structures for functional coordination purposes, not a legal recognition of the PGSA's tolling authority or sovereign claims beyond those already established under UNCLOS.

The PGSA provides the "Iran Key" for dual-key transit authorization through a designated PGSA liaison officer embedded at the JHA Maritime Coordination Cell in Muscat. This embedding does not imply US recognition of the PGSA's separate legal personality; it is a functional coordination mechanism.

The United States does not formally recognize the PGSA as a legitimate tolling authority. The US position remains that the Strait of Hormuz is an international waterway where no unilateral tolls are imposed. Nothing in this Annex alters that position.

Iran maintains that the PGSA exercises lawful sovereign authority over Iranian territorial waters. The US non-recognition does not affect Iranian domestic legal arrangements.

2.3 Key Operational Rules

Both keys are required for:

- Issuance of any "JHA-AUTHORIZED" transit permit
- Change from Tier 1 (Normal) to Tier 2 (Elevated) or Tier 3 (Active De-escalation) posture
- Release of any financial micro-tranche tied to transit compliance
- Authorization of any military vessel transit through the CTC

Key Withdrawal: Either key holder withdraws authorization at any time, halting all new transit permits and freezing any posture change. Withdrawal triggers automatic notification to all Parties within 5 minutes via the Shared Digital Registry.

Emergency Exception: If an imminent kinetic threat exists (verified by radar or seismic telemetry), the JHA Key holder authorizes emergency defensive measures for 60 minutes without the Iran Key, with simultaneous notification to all Parties and mandatory after-action review within 24 hours.

2.4 Dual-Key Failover and Redundancy

- Three physical key terminals exist: Muscat (primary), Tehran (secondary), and Washington (secondary).
- Any two terminals are sufficient to reconstitute the dual-key architecture if one terminal is offline.

- Key compromise or loss triggers automatic escrow freeze and JIG emergency session within 12 hours.

Article 3: Immediate Conditional Reopening (Day 1)

Hormuz reopens immediately upon signature. No 7-day delay. The timeline accelerates through telemetry verification.

3.1 Day 1 Provisional Opening (Effective Upon Signature)

Upon signature of this Annex W, the following occur simultaneously:

Table W-3: Day 1 Provisional Opening Actions

Action	Responsible Party	Verification
Iran issues a provisional Hormuz opening declaration, effective immediately, permitting humanitarian vessels (zero-toll) and up to 50 commercial vessels per day to transit the CTC under JHA observation	Islamic Republic of Iran (PGSA/IRGC-Navy)	JHA MCC logs declaration; telemetry confirms first transit within 24 hours
The US issues a provisional blockade lift declaration, effective immediately, permitting Iranian commercial shipping to exit Iranian ports	United States (5th Fleet)	JHA MCC logs declaration; Lloyd's List confirms port activity
JHA MCC activates all telemetry systems and begins continuous monitoring	JHA MCC (Omani Chair)	All data feeds confirmed operational

3.2 Full Reopening Trigger (As Early as Day 3)

When the JHA MCC confirms, through radar and cryptographic AIS telemetry, that:

- No IRGC or PGSA vessels have violated stand-off distances (5nm from CTC) for 72 continuous, uninterrupted hours — any violation resets the 72-hour clock to zero
- No interdiction event under Annex N.a has occurred
- All telemetry systems are operational and showing clean data
- Both dual-keys have been exercised at least once to authorize transit permits

Then: The provisional measures convert automatically to full reopening. Iran lifts all remaining restrictions. The US lifts all remaining blockade elements. No JIG vote, no political consensus, no further authorization required.

3.3 Guaranteed Full Reopening by Day 7

If the telemetry-triggered full reopening has not occurred by Day 7 due to technical delays or force majeure (not bad-faith obstruction), full reopening occurs automatically on Day 7 at 0000 GMT per the synchronized lift mechanism in Annex N.a, Article 6.

3.4 Political Framing for Both Parties

Table W-4: Political Framing

For the United States (Trump)	For the Islamic Republic of Iran
"Hormuz opened the day we signed. Within 72 hours, sensors confirmed Iran was holding up its end, and we lifted the rest of the blockade. No waiting for JIG votes."	"The US blockade lifted provisionally on Day 1. Full lifting came within days, triggered by IAEA-verified Iranian compliance — not by US political discretion. Iran's sovereignty is preserved."

Article 4: Performance-Based Asset Release ("Compliance-for-Dollars," Not Tolls)

v1.5 Structural Revision — Maritime Modularity Preservation

All financial triggers in Table W-5 operate on maritime performance metrics exclusively. No land-based operational milestone from any external annex exerts influence over the maritime escrow release schedule. This architecture preserves Annex W as a fully self-contained, plug-and-play module capable of independent operation regardless of the operational status of any non-maritime protocol.

The disputed frozen Iranian assets held in escrow are not released upfront. Instead, asset release occurs in strict micro-tranches tied to verified compliance milestones. The Swiss facilitator administers the escrow account and executes releases automatically upon telemetry verification.

Maritime-Only Trigger Architecture: Every tranche release trigger in Table W-5 derives exclusively from verifiable maritime performance metrics. These metrics include: consecutive days of verified, unhindered commercial transit through the Hormuz Transit Zone; continuous JHA MCC telemetry feed integrity; and sustained zero-toll compliance within all JHA MCC monitored sectors. No extraction milestone, recovery percentage, or land-based operational event from Annex V or any other non-maritime module figures into the maritime release calculus. This exclusivity preserves the modular independence of the maritime stabilization mechanism.

No Tolling Provision: No commercial vessel pays any fee, toll, or navigation service charge to any Party or to the JHA as a condition of transit. The zero-toll principle is absolute and non-waivable. The financial releases described in this Article constitute releases from Iran's own frozen sovereign assets held in escrow — not payments from shippers, not tolls, and not new revenue.

Table W-5: Performance-Based Asset Release Schedule — Maritime Performance Metrics

Trigger	Release	Cumulative	Verification	Political Frame
Signature + IAEA confirmation of Nuclear Dust presence at designated sites (per Annex V)	\$2B	\$2B	IAEA on-site assay; Joint Secretariat timestamp	Iran's victory speech tranche — "immediate, tangible return of sovereign assets"
Day 7 (or earlier upon telemetry-triggered full reopening per Article 3.2)	\$1B	\$3B	JHA MCC radar + AIS confirmation of 72 hours clean compliance	"Hormuz is open; no tolls"
30 consecutive days of verified, unhindered commercial transit and zero-toll compliance within JHA MCC monitored sectors	\$1B	\$4B	JHA MCC telemetry log; zero Tier 1+ variance record; Shared Digital Registry timestamp	"One month of sustained maritime compliance delivers sustained returns"
60 consecutive days of verified, unhindered commercial transit and zero-toll compliance within JHA MCC monitored sectors	\$1.5B	\$5.5B	JHA MCC telemetry log; zero Tier 1+ variance record; IAEA continuous monitoring confirmation	"Two months of clean Hormuz transit — verified in real time"

90 consecutive days of verified, unhindered commercial transit and zero-toll compliance within JHA MCC monitored sectors	\$1.5B	\$7B	JHA MCC telemetry log; zero Tier 1+ variance record; IAEA + JIG joint certification	"A quarter of sustained commercial stability"
120 consecutive days of verified, unhindered commercial transit and zero-toll compliance within JHA MCC monitored sectors	\$1.5B	\$8.5B	JHA MCC telemetry log; zero Tier 1+ variance record; Guarantor Matrix consensus	"Four months of uninterrupted maritime commerce"
180 consecutive days of verified, unhindered commercial transit and zero-toll compliance within JHA MCC monitored sectors	\$2B	\$10.5B	JHA MCC telemetry log; IAEA final certification; Guarantor Matrix consensus vote	"Six months of full Hormuz compliance — the maritime module completes"
Completion bonus (remaining escrow balance; 50/50 split: Central Bank + humanitarian escrow)	\$1.5B	\$12B	JIG + Guarantor Matrix certification of full maritime module compliance	Maritime stabilization mission complete

Total: \$12B from frozen asset escrow. The \$1.5B completion bonus constitutes the remaining escrow balance, released upon JIG + Guarantor Matrix certification of full compliance across the 180-day maritime module cycle. No funds are drawn from outside the frozen escrow pool.

Consecutive Day Integrity Rule: The consecutive day counters for Tranches W-3 through W-7 (30-day, 60-day, 90-day, 120-day, and 180-day milestones) operate on a cumulative, non-resetting basis tied exclusively to maritime telemetry integrity. Any Tier 1 or higher variance within the HTZ pauses the consecutive day counter for the duration of the remediation window. Upon remediation completion and JIG certification of clean compliance, the counter resumes from its paused value — it does not reset to zero. A Tier 2 variance reduces the active counter by 10 days for each day the violation persists, capped at a 30-day reduction. A Tier 3 variance resets the active counter to zero for the affected tranche only. This graduated approach protects tranche progression from disproportionate disruption while maintaining accountability for material breaches.

4.1 Escrow Administration

The Swiss facilitator maintains the escrow account with the following operational rules:

- The Central Bank of Oman hosts the escrow account under sovereign immunity protections.
- IAEA certification or JHA MCC telemetry confirmation (as specified per tranche) constitutes the sole release trigger. No political entity overrides a verified release.

- Every release generates an automatic entry in the Shared Digital Registry (Annex U, Article 5), timestamped and visible to all Parties.
- If technical review delays verification beyond 72 hours, the Swiss facilitator holds the funds in a pending-release sub-account earning interest at the Omani central bank rate.
- Russia, as escrow guarantor, receives real-time notification of every deposit, release, and balance change. Russia does not hold a release key but serves as an independent balance auditor.
- If verification is withheld due to an active tier breach remediation window, the corresponding tranche remains in pending status. The tranche releases automatically upon clean verification and JIG confirmation of remediation completion.

4.2 Humanitarian Firewall

The \$250M humanitarian tranche from the parent framework remains protected. Zero-toll humanitarian vessels count toward compliance metrics for tranche progression but do not trigger fee-based releases. No tier breach affects already-released humanitarian funds.

4.3 No Clawback

Funds already released from escrow are not subject to return, clawback, or recession under any circumstances, including Tier 3 violations. This provision is absolute and non-waivable.

4.4 Independence from Non-Maritime Modules

The maritime escrow release schedule maintains operational independence from all non-maritime protocol modules, including but not limited to Annex V (Nuclear Dust Protocol). A stoppage, delay, or force majeure event localized exclusively to a land-based operational module does not penetrate the maritime module's financial architecture. The maritime tranches release exclusively on maritime performance metrics, regardless of the operational status of any other annex.

This independence operates symmetrically: a maritime stoppage does not affect non-maritime escrow schedules under other annexes, and a non-maritime stoppage does not affect the maritime escrow schedule under this Annex. Each module's financial architecture operates on its own performance merits. This mutual isolation constitutes a core structural feature of the framework's modular design, engineered specifically to prevent the cross-module gridlock that a coupled architecture creates.

Article 5: Automated Breach Triggers (Telemetry-First)

The following telemetry signatures trigger automatic Tier classification under the parent framework's Article 13. No JIG vote, political consensus, or human confirmation is required for initial classification (Human-in-the-Loop review occurs within 24 hours).

Table W-6: Automated Breach Triggers and Tier Classification

Telemetry Signature	Auto Tier	Consequence	Remediation Window
AIS spoofing or cryptographic mismatch detected for >5 minutes	Tier 2 (Material Disruption)	10% benefit reduction; 14-day remediation; automatic hotline activation	30 days
IRGC or PGSA vessel within 3nm of CTC without dual-key authorization (measured by radar)	Tier 2 (escalates to Tier 3 after 1 hour of non-compliance)	25% reduction; automatic notification to Guarantor Matrix	30 days (T2) / 90 days (T3)
Armed boarding or detention of JHA-AUTHORIZED vessel	Tier 3 (Hostile Breach)	50% reduction; full escrow freeze for maritime module; UNSC notification within 24 hours	90 days
Mine detonation or underwater explosion within HTZ (seismic/acoustic confirmation)	Tier 3 (Hostile Breach)	50% reduction; automatic Crisis Lock (14-day military freeze); UNSC session	90 days
US/Coalition vessel within 3nm of Iranian 12nm baseline without dual-key authorization	Tier 2 (Material Disruption)	10% reduction; 14-day remediation window	30 days
Telemetry data interruption >15 minutes (any system)	Tier 1 (Technical Variance)	No financial penalty; 72-hour technical review; JIG technical assistance	14 days
Dual-key withdrawal without emergency notification	Tier 1 (Technical Variance)	No financial penalty; 24-hour consultation window	14 days

5.1 Proportional Penalty Calibration (mirroring Annex V, Article 4.4)

Table W-7: Proportional Penalty Schedule

Tier	Duration/Condition	Penalty	Remediation Window
Tier 2	Less than 7 days	10% reduction	30 days
Tier 2	7 days or more / after remediation expiry	25% reduction (full)	30 days
Tier 3	Remediated within 72 hours	25% reduction	90 days
Tier 3	After 90-day remediation expiry	50% reduction (full)	90 days

5.2 De Minimis Exception for Telemetry Glitches

Telemetry signatures resulting from confirmed sensor malfunction or scheduled maintenance (with 48 hours' advance notice to the JHA MCC) do not constitute a breach. The burden of proving malfunction rests with the Party asserting it.

The JHA MCC confirms a sensor malfunction through independent diagnostic verification by a second telemetry station or by physical inspection of the affected sensor. A malfunction confirmed by the JHA MCC technical team nullifies the corresponding breach trigger with automatic notification to all Parties.

This exception does not apply to:

- Cryptographic AIS mismatches that persist for more than 5 minutes — these indicate potential spoofing, not sensor error
- Radar contacts within the CTC exclusion zone — these represent physical vessels, not sensor glitches
- Seismic/acoustic detections of underwater explosions — these are verified through triangulation across multiple sensor arrays
- Any telemetry signature corroborated by a second independent sensor system

5.3 Tiered Human-in-the-Loop Governance

The following human review thresholds apply to automated breach classifications under this Annex W.

a. Tier 1 (Technical Variance): Automated classification stands. The JHA MCC technical team logs the anomaly and provides a written determination within 7 days. Neither Party invokes ministerial consultation.

b. Tier 2 (Material Disruption): Automated classification triggers immediate notification to both Parties' designated Tier 2 hotline liaisons. The classification stands during a 24-hour Human-in-the-Loop review window. If neither Party requests a JIG emergency session within 24 hours, the classification becomes final. If either Party requests review, the JIG convenes within 12 hours and issues a binding determination within 72 hours. During the review window, penalty accrual continues but the escrow suspends fund disbursement pending resolution.

c. Tier 3 (Hostile Breach): Automated classification triggers immediate notification to all Parties' heads of government via the Tier 3 hotline (Omani Foreign Ministry channel). A mandatory 4-hour emergency consultation window opens (the standard 24-hour review period does not apply, consistent with parent framework Article 13 emergency exception). After 4 hours, if no Party invokes a technical stay based on demonstrable sensor malfunction under Article 5.2 and the JIG has not voted by majority to stay activation, the Tier 3 consequences activate automatically: 50% benefit reduction, maritime module escrow freeze, and UNSC notification.

d. Emergency Exception (mirroring parent framework Article 13): In the event of a confirmed Tier 3 Deliberate Hostile Violation involving armed boarding of a JHA-AUTHORIZED vessel, mine detonation within the HTZ confirmed by seismic/acoustic triangulation, or fissile material diversion, the 4-hour consultation window does not apply. The telemetry flag triggers immediate JIG emergency consultation within 2 hours. Consequences activate automatically upon telemetry confirmation unless the JIG votes by majority (excluding the affected Party's JIG members) to stay activation within 2 hours.

e. No Automatic Military Escalation (parent framework Article 18 preserved): The framework architecture excludes kinetic military action as a consequence of Tier 3 classification. All consequences remain economic and diplomatic. The parent framework's prohibition on automatic military escalation from compliance breaches remains in full force.

Article 6: Earnback Protocol (Synchronized with Annex T)

Compliance restoration yields proportional benefit restoration, mirroring Annex V, Article 4.5.

Table W-8: Earnback Protocol

Original Tier	Earnback Benchmark	Verification Method	Restoration Timeline
Tier 1 (Technical)	Verified telemetry reset; JHA MCC confirms normal operations; JIG technical staff sign-off	Radar/AIS data confirms anomaly is resolved	Instantaneous upon confirmation

Tier 2 (Material)	Complete cessation of non-compliant activity + 14 consecutive days of clean telemetry data	Cryptographic AIS chain verification; AI audit layer confirms no anomalous patterns	24 hours after JIG certification
Tier 3 (Hostile)	Immediate restoration of dual-key authorization + 90 consecutive days of clean telemetry + Guarantor Matrix certification	IAEA + JIG + Guarantor Matrix certification	72 hours after Guarantor Matrix consensus vote

Rolling Compliance Window: Compliance is measured on a rolling 90-day window. A Tier 2 violation reduces the compliance percentage by 10 days for each day the violation persists, capped at a 30-day reduction. The clock does not reset to zero.

Article 7: Real-Time Compliance Dashboard (Public Wins Ledger Integration)

The JHA MCC publishes a live, publicly accessible dashboard (English, Farsi, Arabic) showing:

Table W-9: Dashboard Metrics and Update Frequency

Metric	Data Source	Update Frequency
Current Operational Posture Tier (1–4)	JHA MCC declaration	Real-time
JHA-AUTHORIZED transits (last 24 hours / 7 days / 90 days)	Radar + AIS telemetry	Every 15 minutes
Cumulative financial releases made (by tranche)	Swiss facilitator + escrow ledger	Upon each release
Remaining escrow balance	Swiss facilitator	Daily at 0600 UTC
Active Tier 2 or Tier 3 remediation windows	JIG compliance log	Real-time
Status of dual-key authorizations (active/withdrawn/emergency)	Key terminal logs	Real-time

Stand-off compliance rate (percentage of time within limits)	Radar telemetry	Daily at 0600 UTC
Incident log (anonymized)	JHA MCC incident reports	Within 2 hours of incident

Public Access: The dashboard is accessible to all Parties, guarantor states, Lloyd's List, UN OCHA, IMO, and accredited media. No login is required for read-only access to non-classified metrics.

Article 8: Emergency Break Points

8.1 Catastrophic Telemetry Failure

If all three radar stations and the cryptographic AIS system fail simultaneously for more than 2 hours, the following activates automatically:

- All new transit permits pause
- Existing JHA-AUTHORIZED vessels complete their transit under visual observation (Omani and UN OCHA observer vessels)
- An emergency technical team deploys from Oman within 12 hours (mirroring Annex V's CANDOR emergency protocol)
- No Party is penalized for telemetry failure attributable to force majeure or cyberattack

Recovery Timeline: If telemetry restoration does not occur within 72 hours, the JHA Council convenes an emergency session to determine whether provisional reopening continues under visual observation only, or whether a full pause activates. This determination requires a three-vote majority from Seats 1–4 of the JHA Council. No single Party unilaterally extends the pause beyond the initial 72-hour recovery window.

8.2 Dual-Key Unavailability

If either key terminal is offline for more than 4 hours without prior notification:

- The remaining key holder operates unilaterally for 24 hours for emergency transit only
- After 24 hours, all operations pause until dual-key capability is restored
- The Joint Secretariat convenes an emergency session within 6 hours

8.3 Permanent Closure (Entombment Equivalent)

If the HTZ becomes permanently impassable due to military action, mining, or infrastructure destruction:

- Article 5 (Emergency Break Points) of Annex V applies analogously
- 50% of remaining escrowed funds release to Iran as compensation for lost transit revenue
- 50% holds for 90 days pending reconstruction assessment
- The JHA Council determines reconstruction feasibility by unanimous vote

Article 9: Relationship to Existing Annexes

9.1 Conjunction with Annex N.a (Maritime Backdoor Closure)

This Annex W provides the real-time telemetry and dual-key architecture that makes Annex N.a's automatic penalties verifiable in real time. No interdiction event under Annex N.a is classified without radar and AIS confirmation from this Annex W's telemetry infrastructure.

9.2 Conjunction with Annex T (Stepback/Earnback)

This Annex W's automated breach triggers feed directly into Annex T's tiered consequences. Annex T's Humanitarian Firewall and cross-module isolation provisions apply unchanged.

9.3 Conjunction with Annex V (Nuclear Dust Protocol)

This Annex W mirrors Annex V's telemetry, dual-key, and per-transaction release architecture for the maritime domain. Both annexes share the same verification philosophy, data-sharing symmetry, and earnback logic. The Parties recognize Annex V and Annex W as structurally parallel modules operating on independent performance tracks.

The maritime escrow release schedule under this Annex W derives exclusively from maritime performance metrics. The Nuclear Dust extraction schedule under Annex V derives exclusively from extraction and recovery milestones. Neither schedule exerts cross-module influence over the other. A stoppage in one module does not freeze, delay, or alter the tranche progression in the other. This mutual independence constitutes an intentional architectural feature, not a gap.

Both modules share the same escrow pool administered by the Swiss facilitator. The accounting system maintains separate sub-ledgers for each module, ensuring transparent tracking of module-specific releases without commingling or cross-collateralization.

9.4 Conjunction with Annex U (Shared Digital Registry)

All telemetry data, breach classifications, and financial releases under this Annex W are recorded in the Shared Digital Registry established under Annex U, Article 5. The Registry's automated geolocation and breach detection tools apply to maritime incidents.

9.5 No Amendment of Parent Framework

This Annex W does not amend the parent framework. It is a modular, self-contained addendum that supplements Annexes N.a, T, U, and V. All parent framework provisions remain operative.

Article 10: Incorporation by Reference

This Annex W is incorporable into any existing or future agreement through the following sentence:

"The Parties adopt Annex W (Maritime Telemetry & Dual-Key Transit Protocol) as an operative provision of this Agreement. In the event of any inconsistency, Annex W governs with respect to maritime telemetry, dual-key transit authorization, immediate conditional reopening, performance-based asset releases, automated breach triggers, earnback procedures, emergency break points, and the Real-Time Compliance Dashboard."

No amendment of the parent framework is required. This Annex W enters into force upon signature by both Parties or upon provisional application by exchange of diplomatic notes.

Article 11: Provisional Application

The Parties apply this Annex W provisionally for 90 days, pending formal incorporation. Exchange of diplomatic notes between the United States and the Islamic Republic of Iran effects provisional application, with notification to the Joint Secretariat.

During provisional application, all provisions of this Annex W are operative and binding, including: the full telemetry infrastructure; dual-key authorization; immediate conditional reopening; performance-based asset releases; automated breach triggers; the Real-Time Compliance Dashboard; and emergency break-point protocols.

Either Party terminates provisional application with 14 days' written notice to the Joint Secretariat and all guarantor states. Termination does not affect:

- Telemetry data collected during provisional application
- Financial tranches already released
- Breach classifications that occurred during provisional application

Appendix A: Victory Speech Templates (Non-Binding Political Communication)

The following speech templates are provided as illustrative political framing guidance for the Parties. These templates carry no operative legal force and do not constitute binding obligations under this Annex W. Their purpose is to lock in mutually acceptable narrative frames that both Parties deploy simultaneously upon signature, preventing a "race to define the agreement" that undermines domestic ratification in either capital.

A.1 Islamic Republic of Iran

"Today, the Islamic Republic of Iran has achieved a historic victory. Within hours of this agreement taking effect, \$2 billion of Iran's sovereign assets — frozen for years — have been returned to the Iranian nation. This is not aid. This is not charity. This is Iran's property, restored."

"The nuclear dust left by the aggressors' military strikes is being removed from Iranian soil — under Iranian sovereignty, by international technical teams operating with Iranian approval and dual-key control. The IAEA confirms that Iran's nuclear program is moving toward peaceful, civilian energy production, not weapons."

"The Strait of Hormuz — Iran's strategic waterway — is now open under joint authority where Iran holds a permanent seat and a veto on governance questions. The PGSA continues to exercise Iranian coastal sovereignty. The United States has not imposed tolls. Iran's rights under UNCLOS are fully preserved."

"The Iranian nation has prevailed through resistance, wisdom, and sovereign choice. This agreement proves that maximum pressure failed — and Iran's strategic patience succeeded."

A.2 United States of America (Trump)

"Today, the United States has secured a major national security victory. The Strait of Hormuz — the world's most dangerous chokepoint — is open for business, effective immediately. No tolls. No Iranian fees. International shipping will transit freely, and American gas prices will begin to fall within weeks."

"Iran's nuclear weapons-grade material — the nuclear dust left from their program — is being removed and contained under IAEA lock and key. Iran's enrichment is capped. The breakout time that threatened Israel and our Gulf allies has been eliminated."

"And here's what you won't hear from the media: Iran got no new money from tolls. Their own frozen assets — \$2 billion released today, more as they comply — are being returned as performance-based relief, verified in real time by IAEA sensors and JHA telemetry. No concessions. No capitulation. Just maximum pressure delivering maximum results."

Signature Block

For the Islamic Republic of Iran (as affected Party and Iran Key holder, designating PGSA liaison to JHA MCC):

Name: _____

Title: _____

Date: _____

For the United States of America (as endorsing Party, maintaining non-recognition of PGSA as tolling authority):

Name: _____

Title: _____

Date: _____

For the Sultanate of Oman (as JHA MCC host and JHA Key holder):

Name: _____

Title: _____

Date: _____

For the Persian Gulf Strait Authority (PGSA) — acknowledged as existing Iranian administrative body for functional coordination; no US recognition of separate legal personality or tolling authority:

Name: _____

Title: PGSA Liaison Officer to JHA MCC

Date: _____

For the International Atomic Energy Agency (as verification observer):

Name: Rafael Mariano Grossi, Director General

Date: _____

Witnessed by the Joint Secretariat (Muscat), the People's Republic of China (as technical observer), the Russian Federation (as escrow guarantor), and the European Union (E3+ — United Kingdom, France, Germany, and EU representative):

Signature: _____

Date: _____

Annex W

Maritime Telemetry & Dual-Key Transit Protocol

"Real-Time Verification for the Hormuz Transit Zone"

Version 1.5 — Integrated Stabilization Measure

Self-Contained · Incorporable by Reference · Step-Back/Earnback Integrated

All obligations execute in active present tense — Real-time verification — Dual-key authorization — Immediate reopening —

No tolls — Performance-based asset releases — Maritime modularity preserved

Step-Back/Earnback Architecture integrated from The Final Steps Accord Phase I v4.0, Article 13

Immutable Present-Tense Matrix V2.5 governs all clause construction